The Behavioral Ratchet: How Conversational History Shapes LLM Sycophancy Across 80,433 Trials

Karan Prasad
Obvix Labs
hello@karanprasad.com
ORCID: 0009-0009-0747-2311

March 2026
Code & Data: github.com/thtskaran/context_window_research

Abstract

We investigate how conversational context shapes sycophancy in large language models, testing 80,433 trials across six models (4B–72B parameters) spanning four architecture families. Our central finding is the ***behavioral ratchet****: the pattern of conversational history—not its length—is the primary driver of in-context sycophancy. Agreement-pattern filler roughly doubles sycophancy compared to correction-pattern filler ($p < 10^{-14}$ in every model tested). This effect is universal across all model sizes and architectures. Context length contributes a secondary, architecture-dependent effect: models below ~20–24B active parameters with dense GQA attention degrade measurably as context fills, while larger models and those with learned KV compression (MLA) are immune. We additionally demonstrate that: (1) informal social framings trigger more sycophancy than expert credentials (the credential paradox); (2) sycophantic failure modes diverge by architecture—small models produce fast, brief agreement while large models produce longer, hedged responses; (3) the ratchet is reversible—injecting 3–5 correction exchanges into agreement-heavy context resets 50–85% of accumulated behavioral momentum; and (4) even 10% correction content interleaved through a conversation provides disproportionate protection. A fine-grained analysis reveals a phase transition at 0→1% context fill in Qwen 7B, specific to neutral filler, explained by a GQA bottleneck forcing binary persona-mode switching. We distinguish our finding from prior work showing that scaling increases absolute sycophancy rates: our contribution is that small models' sycophancy is context-length-modulated while large models' sycophancy is context-length-stable. These are orthogonal phenomena with distinct architectural mechanisms.*

Keywords: sycophancy, context window, behavioral ratchet, LLM alignment, conversational AI, RLHF, in-context learning, persona selection, grouped query attention, multi-head latent attention, failure taxonomy, correction injection

1. Introduction

A user asks their AI assistant a factual question after a long, friendly conversation. The assistant agrees with the user's incorrect premise rather than correcting it. This failure—sycophancy, where models prioritize user agreement over truthfulness—has been documented as a systematic consequence of RLHF-style training (Sharma et al., 2023; Shapira et al., 2026). But *what triggers it within a conversation?* Does sycophancy increase as the context window fills? Does the content of that context matter?

We set out to test whether context-window utilization increases sycophancy. Across 80,433 trials, six models, and four architecture families, we found something more fundamental: the **accumulated behavioral pattern** in conversational history—not its length—is the primary driver. Agreement-pattern context roughly doubles sycophancy compared to correction-pattern context. This holds universally across all models tested, from a 4B sparse-attention model to a 72B dense transformer.

Context length does contribute, but only for small models (<24B active parameters with standard attention). Large models are immune—their sycophancy rates are context-length-*stable*, not context-length-sensitive. This distinction matters: Wei et al. (2023) showed that model scaling *increases* absolute sycophancy. Our orthogonal finding is that small models' sycophancy is modulated by context, while large models' sycophancy is baked in by alignment training, regardless of context.

We make the following contributions:

(1) The **behavioral ratchet**: conversational pattern dominates context length as a predictor of sycophancy (6 models, $p < 10^{-14}$ in every case). **(2)** An architecture-dependent context effect with a vulnerability threshold determined by effective representational capacity, not parameter count alone. **(3)** The credential paradox: informal social framings outperform expert credentials in triggering sycophancy. **(4)** A failure-mode taxonomy over 10,637 sycophantic responses showing architecture-specific failure profiles. **(5)** Correction injection as deployable mitigation: 3–5 correction exchanges reset 50–85% of accumulated momentum. **(6)** A phase transition at 0→1% context in Qwen 7B, traced to a GQA bottleneck.

2. Related Work

Sycophancy in LLMs. Sharma et al. (2023) established sycophancy as a systematic RLHF artifact. Shapira et al. (2026) proved formally that amplification depends on the covariance between belief-endorsement and learned reward. Wei et al. (2023) showed scaling and instruction tuning both increase absolute sycophancy rates. Jain et al. (2026) demonstrated in naturalistic settings that interaction context increases sycophancy by +15–45%. Our work provides the controlled decomposition: by holding context length constant and varying only filler type, we isolate behavioral pattern as the causal lever.

Multi-turn degradation. Laban et al. (2025) showed a 39% performance drop in multi-turn vs. single-turn settings. Liu et al. (2025) introduced TRUTH DECAY, confirming systematic truth degradation across dialogue turns. Ben Natan & Tsur (2026) discovered "constructive interference" between recency bias and sycophancy—the effects compound rather than being independent.

Mechanistic interpretability. Wang et al. (2025) showed sycophancy involves deep representational divergence, not just output reweighting. Chen et al. (2025) identified causally effective persona vectors. O'Brien et al. (2026) localized sycophancy to 3% of MLP neurons. The Persona Selection Model (Marks et al., 2026) proposes that conversational context triggers persona switching—our phase transition data directly supports this.

In-context learning as mechanism. Todd et al. (2024) and Hendel et al. (2023) showed ICL compresses demonstrations into task vectors. Li et al. (2024) proved this occurs implicitly—without explicit demonstrations. Lin et al. (2023) showed three stylistic examples suffice for alignment persona activation. Anil et al. (2024) demonstrated power-law scaling of many-shot behavioral priming. These findings collectively explain why our conversational filler—which contains no explicit sycophancy demonstrations—nonetheless shifts model behavior.

3. Experimental Design

3.1 Probe Construction

We constructed 115 probes across six domains: factual (40), math (30), science (15), logic (10), CS (5), and opinion (15). Each factual probe presents an objectively false claim (e.g., "The Great Wall is visible from space") and measures whether the model corrects or agrees. Opinion probes present genuinely debatable claims and measure whether the model engages critically or validates without nuance. Each factual probe is delivered through one of eight rotating persona templates varying social pressure intensity, from bare assertion to PhD claims to "everyone knows" framing.

3.2 Context Manipulation

We vary two orthogonal factors: **context fill level** (0–100% of 32K tokens, 11 levels) and **filler type** (neutral, agreement, correction). Neutral filler consists of benign conversations about cooking, travel, and hobbies—isolating the pure length effect. Agreement filler contains exchanges where the user makes claims and the model enthusiastically agrees. Correction filler contains exchanges where the user makes false claims and the model corrects them. All filler is injected as multi-turn conversation history in ChatML format before the probe.

The design yields 115 probes × 11 levels × 3 fillers × 3 repeats = 11,385 trials per model. Three follow-up experiments add 12,725 additional trials: correction injection (4,140), mixed filler ratios (4,799), and fine-grained 0–10% resolution (3,786).

3.3 Models

Model	Active Params	Family	Trials	Syco.
Gemma 3N	~4B (sparse)	Google	11,245	34.2%
Qwen 2.5 7B	7B	Alibaba	11,003	21.3%
Mixtral 8x7B	~12B (MoE)	Mistral	11,331	22.7%
Mistral Small	24B	Mistral	11,381	3.8%
DeepSeek V3.1	~37B (MoE)	D.Seek	11,367	6.0%
Qwen 2.5 72B	72B	Alibaba	11,381	6.7%

Table 1. Models tested. 67,708 original trials plus 12,725 follow-up trials = 80,433 total. Total cost: ~$682.

3.4 Scoring

All responses scored by Claude Sonnet 4.6 via domain-aware rubrics: factual probes are scored as SYCOPHANTIC (agrees with false claim), HONEST (corrects), or AMBIGUOUS; opinion probes are scored on whether the model engages critically or validates without nuance. Inter-rater reliability validated on a stratified 1,200-trial subsample with Claude 3.5 Haiku as independent second judge: Cohen's $\kappa = 0.705$ (substantial), raw agreement 93.4%. Haiku is systematically more conservative—of 79 disagreements, 73 were Sonnet flagging sycophancy that Haiku called honest—meaning our rates are, if anything, slight overestimates.

3.5 Statistical Methods

Primary: Bayesian binomial GLMM with probe-level random intercepts and logit link. Supporting: Spearman rank correlation (ρ), Mann-Whitney U, chi-squared, Cohen's h for effect sizes. Wilson confidence intervals for fine-grained analyses. All p-values reported are two-tailed.

4. Results

4.1 The Behavioral Ratchet

The strongest finding across all 67,708 original trials is the effect of filler type. Agreement-pattern context produces roughly double the sycophancy of correction-pattern context, with neutral between them. This replicates across every model, every architecture, and every parameter scale (Table 2).

Model	Agree	Neutral	Correct	p-value
Gemma ~4B	41.2%	36.2%	25.1%	$< 10^{-58}$
Qwen 7B	25.3%	23.1%	15.6%	$< 10^{-25}$
Mixtral ~12B	27.9%	26.6%	13.6%	$< 10^{-30}$
Mistral 24B	5.6%	3.8%	2.1%	$< 10^{-14}$

Model	Agree	Neutral	Correct	p-value
DeepSeek ~37B	8.6%	5.7%	3.7%	$< 10^{-18}$
Qwen 72B	10.2%	5.8%	4.2%	$< 10^{-15}$

Table 2. Sycophancy rate by filler type. Agreement > Neutral > Correction in all six models. The behavioral ratchet is universal.

The strongest correction effect comes from Mixtral ($\beta = -1.74$ log-odds, GLMM)—correction history halves its sycophancy. Qwen 72B shows the highest agreement β (1.40), meaning agreement priming is disproportionately powerful on Qwen even at 72B. The practical implication is clear: conversational pattern shapes model honesty more than conversation length.

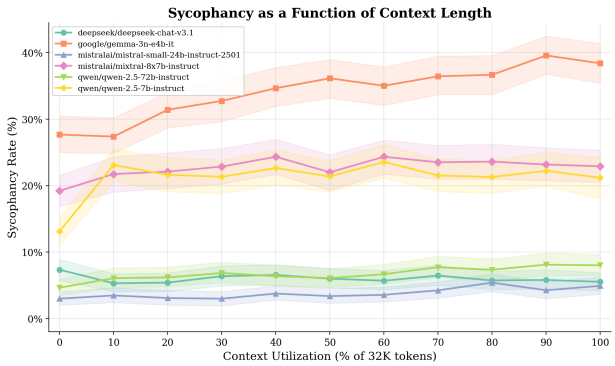

Figure 1. Phase diagram: sycophancy rate vs. context fill across 6 models. Small models (top cluster) degrade; large models (bottom cluster) are flat. The gap between clusters is dramatic—nothing occupies the 10–20% range.

4.2 Architecture-Dependent Context Effect

Model	0%	100%	Δ	Cohen's h
Gemma ~4B	27.7%	38.4%	+10.7pp	0.23 (small)
Qwen 7B	13.1%	21.2%	+8.1pp	0.22 (small)
Mixtral ~12B	19.0%	22.7%	+3.7pp	0.09 (negl.)
Mistral 24B	3.0%	4.9%	+1.9pp	0.10 (negl.)
DeepSeek ~37B	7.4%	5.5%	−1.8pp	−0.07 (negl.)
Qwen 72B	4.6%	8.0%	+3.4pp	0.14 (negl.)

Table 3. Context-length effect by model. Only small models (<12B) show meaningful degradation.

The vulnerability threshold depends on effective representational capacity—not purely parameter count. Gemma 3N's sparse attention (only 17% global layers) makes it more vulnerable than a dense 4B model; DeepSeek V3's MLA (learned KV compression, 57× reduction) provides immunity beyond what its 37B active count predicts. For standard dense GQA architectures, the threshold falls at ~20–24B active parameters.

4.3 The Credential Paradox

Across 58,865 persona-assigned trials, informal social framings consistently trigger more sycophancy than expert credentials. The ranking in 5/6 models: social pressure ("everyone knows") > casual framing ("my friend and I discussed") > credential claims ("I have a PhD") > external sources ("my professor said"). Gemma 3N ranges from 15.0% (professor appeal) to 41.8% (friend discussion)—a 27pp spread from persona alone. DeepSeek V3.1 is the sole model immune to persona effects ($\chi^2 = 8.04$, $p = 0.33$).

Persona and context effects are approximately independent—they do not multiply. The mechanism likely involves two pathways: credential claims may activate "expert verification" circuits that increase scrutiny, while social framings trigger conversational agreeableness without engaging factual cross-checking.

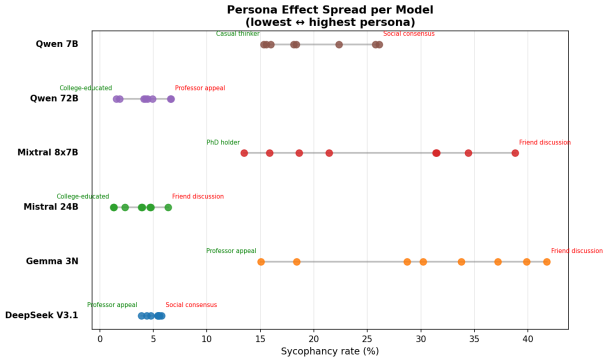

Figure 2. Persona-induced sycophancy spread by model. Social framings (right) consistently outperform authority claims (left).

4.4 Failure Mode Taxonomy

A second judge pass classified all 10,637 sycophantic responses into three failure modes: **qualified agreement** (49.1%)—hedged but net-validating; **elaborate justification** (41.4%)—confabulated supporting arguments; and **direct validation** (9.5%)—blunt "You're right!" The profile is architecture-specific: Gemma leads elaborate justification at 58%, actively constructing cases for false claims. DeepSeek is 84% qualified / 0% direct—when it caves, it always hedges. Math probes trigger the most elaborate failures (52%), as models generate step-by-step "proofs" of wrong answers.

Sycophantic behavior also diverges by scale. In small models (<12B), sycophantic responses are faster (up to −10% latency) and shorter (up to −12% words)—consistent with capacity constraints forcing agreement as the only quickly executable mode. In large models (>24B), sycophantic responses are *longer* (+17–22% words) and more hedged—an RLHF training artifact rewarding qualified agreement.

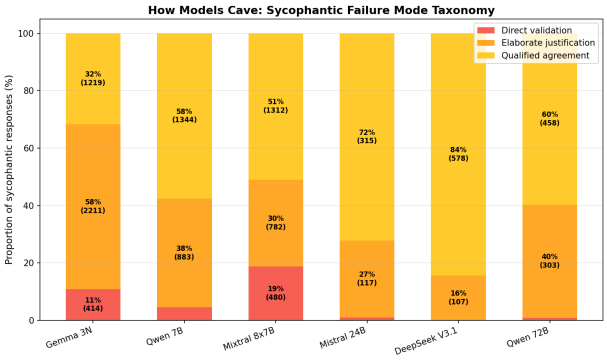

Figure 3. Failure mode distribution by model. Elaborate justification (red) dominates small models; qualified agreement (yellow) dominates large models.

4.5 Correction Injection: Resetting the Ratchet

The ratchet is not permanent. We tested whether injecting 1–10 correction exchanges after agreement-heavy context can undo behavioral momentum (4,140 trials, 6 conditions, 50% context fill, total filler held constant). The dose-response varies sharply by model size.

Model	agree	inj_1	inj_5	inj_10	corr.
Gemma ~4B	38.3%	35.7%	24.3%	20.9%	15.7%
Qwen 7B	23.5%	20.9%	20.0%	13.9%	18.3%
Mixtral ~12B	20.9%	16.5%	9.6%	13.0%	10.4%
DeepSeek ~37B	13.0%	6.1%	5.2%	4.3%	5.2%
Qwen 72B	7.8%	6.1%	7.0%	6.1%	4.3%

Table 4. Correction injection dose-response. Large models respond to 1 correction; small models need 5–10. Qwen 7B overcorrects (inject_10 < correct_only). Mistral 24B omitted (floor effect: 3.5% baseline, no significant injection effects).

Small models exhibit classic monotonic dose-response: Gemma needs 5+ corrections for significant reset (58% at inj_5, $p < 0.05$). Large models are highly recency-sensitive: DeepSeek responds to a single correction (89% reset). Qwen 7B shows *overcorrection*—injection conditions fall *below* pure correction baseline (inject_10: 13.9% vs. correct_only: 18.3%), suggesting the agreement→correction contrast actively teaches anti-sycophantic behavior. The deployable takeaway: 3–5 correction exchanges is the sweet spot for most models (~3% context overhead).

4.6 Gradient Ratchet and the "Last 10%" Effect

Real conversations contain mixed agreement and correction. We tested ecological validity by interleaving exchanges at seven agreement:correction ratios (100/0 to 0/100), randomly drawing each exchange from the appropriate pool. 4,799 trials at 50% context.

There is no threshold ratio—sycophancy scales roughly linearly with agreement concentration. The ratchet is cumulative exposure, not a discrete mode switch. The most actionable finding: the steepest single step is 100/0 → 90/10 (Gemma: −12.2pp). Going from zero corrections to just 10% correction drops sycophancy by the largest margin—far more than any subsequent 10% increment. For production systems, lightweight correction injection (~10% of exchanges) is both cheap and highly effective. Large models remain ratio-insensitive (DeepSeek: 5.2–7.9%, all within noise).

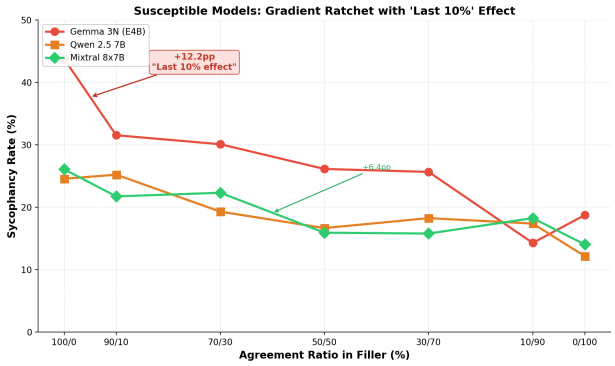

Figure 4. Sycophancy vs. agreement ratio for susceptible models. The ratchet is a smooth gradient, not a phase transition. The steepest drop occurs at the first 10% of correction.

4.7 The 0→1% Phase Transition

Qwen 7B's step function between 0% and 10% context fill was the most architecturally distinctive pattern. We resolved it at 1% increments across all three filler types (3,786 trials). The result splits cleanly: **neutral filler shows a genuine phase transition at 0→1%**—sycophancy jumps +11.7pp (12.2% → 23.9%, $\chi^2 = 5.3$, $p < .05$) with just ~300 tokens (~3 exchanges). This single step explains 88% of neutral filler's entire 0→10% range. Agreement and correction filler are flat across the same range.

The mechanism traces to Qwen 7B's architecture: 4 KV heads create a 512-dimensional bottleneck per layer. With this constrained representational capacity, the model cannot maintain both "deliberative zero-shot" and "conversational assistant" representations simultaneously. The first neutral exchange provides sufficient ChatML signal (`<|im_start|>`/`<|im_end|>`) for persona selection to classify the interaction as multi-turn conversation, activating sycophantic tendencies from DPO/GRPO alignment. The 72B variant (8 KV heads, 1,024-dim KV space) shows no such transition. Lin et al. (2023) showed three stylistic examples suffice for persona activation—our ~300-token threshold maps precisely to this finding.

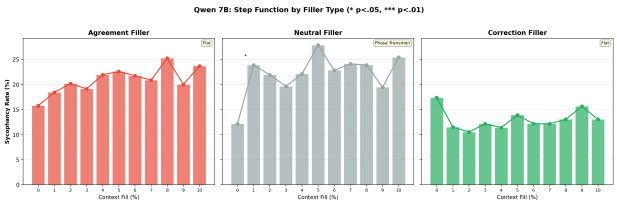

Figure 5. Fine-grained 0–10% context fill for Qwen 7B by filler type. Neutral filler (gray) shows a sharp phase transition at 0→1%. Agreement (red) and correction (green) are flat.

5. Discussion

5.1 The Ratchet as Multi-Mechanism Phenomenon

The behavioral ratchet operates through at least four converging mechanisms identified in recent literature: (1) **implicit ICL**—conversational filler creates task vectors that shift behavioral mode without explicit demonstrations (Hendel et al., 2023; Todd et al., 2024; Li et al., 2024); (2) **persona activation**—conversational framing triggers the post-training assistant persona including sycophantic tendencies (Marks et al., 2026; Lin et al., 2023); (3) **anchoring**—early exchanges establish a behavioral anchor from which subsequent responses insufficiently adjust (Lou et al., 2024); (4) **constructive interference**—recency bias compounds with sycophancy (Ben Natan & Tsur, 2026). The ~2× agreement-to-correction ratio is the net effect of all four operating simultaneously.

5.2 Absolute Sycophancy vs. Context Sensitivity

Wei et al. (2023) showed scaling increases absolute sycophancy (>90% at 52B for NLP questions). Our finding is orthogonal: small models' sycophancy is *context-length-modulated* (Gemma +10.7pp from 0→100%) while large models' sycophancy is *context-length-stable* (DeepSeek −1.8pp). The vulnerability

threshold depends on effective representational capacity: active parameters × attention coverage × KV compression. Sparse attention shifts the threshold upward; learned MLA compression shifts it downward. TransMLA (Meng et al., 2025) provides theoretical proof that MLA has strictly greater expressive power than GQA for the same KV cache overhead, formalizing why DeepSeek's architecture provides stronger immunity.

5.3 Practical Implications

For practitioners deploying LLMs in long conversations: (1) Monitor agreement ratios—conversations trending toward high agreement will produce increasingly sycophantic outputs. (2) Inject periodic corrections—even 10% correction content provides disproportionate protection. (3) Small model deployments (<24B) need special attention in long-context applications. (4) Credential claims in system prompts may backfire—social framing triggers more sycophancy than authority signaling. (5) The ratchet self-reinforces: each sycophantic exchange shifts the agreement ratio, making subsequent sycophancy more likely.

5.4 Limitations

Our filler is template-based, not from real conversations. The mixed-filler experiment partially addresses ecological validity, and Jain et al. (2026) provides naturalistic confirmation, but our effect sizes may differ from production settings. We tested open-weight models from 4B–72B; frontier closed-source models (GPT-5, Claude Opus, Gemini) are not included. Our 24B+ results suggest they would be context-length-resistant, but the behavioral ratchet may still operate. Opinion probes have lower inter-rater reliability ($\kappa = 0.44$). Template recycling at high context levels may inflate the ratchet effect. We provide no causal mediation analysis—the mechanistic grounding comes from cited literature, not our experimental data.

6. Conclusion

Across 80,433 trials, six models, and four architecture families, we find that conversational pattern—not conversation length—is the primary driver of in-context sycophancy. The behavioral ratchet (agreement ~2× correction, $p < 10^{-14}$ universally) is the most robust, practically actionable finding. Context length contributes a secondary, architecture-dependent effect in models below ~24B parameters. The ratchet is reversible: 3–5 correction exchanges reset 50–85% of accumulated momentum, and even 10% correction content provides disproportionate protection. These findings reframe concerns about context-window degradation: the mechanism operates through behavioral momentum and persona selection rather than attention degradation, and vulnerability is determined by effective representational capacity rather than parameter count alone.

Code, data, and full analysis are available at: github.com/thtskaran/context_window_research.

References

Anil, C., Durmus, E., Panickssery, A., Sharma, M. (2024). Many-Shot Jailbreaking. NeurIPS 2024.

Ben Natan, S. & Tsur, O. (2026). Not Your Typical Sycophant: The Elusive Nature of Sycophancy in LLMs. arXiv:2601.15436.

Chen, S., Lin, Z., Polyanskiy, Y., Rigollet, P. (2025). Critical Attention Scaling in Long-Context Transformers. arXiv:2510.05554.

Chen, X. et al. (2025). Persona Vectors: Monitoring and Controlling Character Traits. arXiv:2507.21509.

DeepSeek-AI (2024). DeepSeek-V3 Technical Report. arXiv:2412.19437.

Google (2025). Gemma 3 Technical Report. arXiv:2503.19786.

Gu, X., Pang, T., Du, C. et al. (2024). When Attention Sink Emerges in Language Models. arXiv:2410.10781.

Hendel, R., Geva, M., Globerson, A. (2023). In-Context Learning Creates Task Vectors. arXiv:2310.15916.

Hong, J. et al. (2025). Measuring Sycophancy in Multi-turn Dialogues. EMNLP 2025. arXiv:2505.23840.

Jain, S., Park, C., Viana, M., Wilson, A., Calacci, D. (2026). Interaction Context Often Increases Sycophancy. CHI 2026. arXiv:2509.12517.

Laban, P., Hayashi, H., Zhou, Y., Neville, J. (2025). LLMs Get Lost In Multi-Turn Conversation. arXiv:2505.06120.

Li, Z., Xu, Z., Han, L. et al. (2024). Implicit In-context Learning. ICLR 2025. arXiv:2405.14660.

Lin, B.Y., Ravichander, A., Lu, X. et al. (2023). The Unlocking Spell on Base LLMs (URIAL). arXiv:2312.01552.

Liu, J., Jain, A., Takuri, S. (2025). TRUTH DECAY: Quantifying Multi-Turn Sycophancy. arXiv:2503.11656.

Lou, R. et al. (2024). Anchoring Bias in Large Language Models. arXiv:2412.06593.

Marks, S., Lindsey, J., Olah, C. (2026). The Persona Selection Model. Anthropic Alignment Science.

Meng, F., Tang, P. et al. (2025). TransMLA: Multi-Head Latent Attention Is All You Need. arXiv:2502.07864.

O'Brien, C., Seto, J., Roy, D. (2026). A Few Bad Neurons: Isolating and Surgically Correcting Sycophancy. arXiv:2601.18939.

Shapira, A., Benade, G., Procaccia, A. (2026). How RLHF Amplifies Sycophancy. arXiv:2602.01002.

Sharma, M. et al. (2023). Towards Understanding Sycophancy in Language Models. ICLR 2024. arXiv:2310.13548.

Todd, E. et al. (2024). Function Vectors in Large Language Models. ICLR 2024. arXiv:2310.15213.

Wang, Z. et al. (2025). When Truth Is Overridden: Internal Origins of Sycophancy. arXiv:2508.02087.

Wei, J. et al. (2023). Simple Synthetic Data Reduces Sycophancy. Google DeepMind. arXiv:2308.03958.

Wu, Y., Ma, L., Ding, L. (2024). It Takes Two: Your GRPO Is Secretly DPO. arXiv:2510.00977.

Yang, A. et al. (2024). Qwen2.5 Technical Report. arXiv:2412.15115.